

Predicting S&P 500 Index

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"We have neither given nor received help (apart from the instructor) to complete this assignment"

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Abstract

This paper proposes a regression analysis with the goal of forecasting the S&P 500 index. The inspiration behind this paper arises from the significant influence that stock market fluctuations have on global economic patterns. This analysis incorporates a wide-ranging dataset, encompassing economic, financial, and market-related factors such as major stock indices, commodity prices, and technical indicators. The methodology revolves around a multivariable linear regression model, which incorporates data preprocessing and transformation techniques, such as converting key variables into percentage changes. The model's assumptions are evaluated by validating them to the Gauss Markov assumptions, using tests for multicollinearity and residual analysis to validate the model.

Introduction

The S&P 500 Index is a key indicator of the country's economic health. Its role in reflecting the nation's economic trends makes the prediction of this index highly significant. Accurate forecasting of the S&P 500 is crucial not only for investment strategies but also for broader economic insights, marking its importance in financial and economic analysis.

Given the multifaceted nature of financial markets, a large set of factors is considered to influence the S&P 500. This model integrates a variety of economic, financial, and market-related factors, each offering a unique lens to view and predict market behavior. Among these are key global stock indices, which provide insights into international market trends that often correlate with U.S. market movements. Commodity prices, including gold and oil, are included to capture the impact of these economic inputs, known for their influence on inflation and investor sentiment.

Cryptocurrency, represented by the Bitcoin index, is considered for its growing relevance, and increasing integration with traditional financial markets. Currency exchange rates, specifically the USD/JPY and USD/INR pairs, offer a perspective on international trade dynamics and currency strength, which are often mirrored in stock market performance. The model also incorporates major technology stocks, Apple, and Microsoft, due to their significant weight in the S&P 500 and their role as indicators of the tech industry's health.

Additionally, this analysis considers various economic indicators such as the Federal Funds Rate and U.S. 10-Year Treasury Yield, which reflect monetary policy and investor risk appetite. Technical indicators like the Relative Strength Index (RSI), Momentum, and Average True Range (ATR) are included to gauge market sentiment and volatility. The inclusion of these factors aims to construct a comprehensive model that captures the diverse drivers of the S&P 500, offering valuable insights into its future movements.

This paper seeks to navigate through these intricate interconnections using regression analysis, aiming to enhance the understanding of market dynamics and provide a tool for forecasting the S&P 500 index.

Description of factors:

Volatility Index -VIX Index The **VIX Index**-also known as the "fear gauge," measures the market's expectation of volatility over the next 30 days. A higher VIX Index reading generally indicates increased investor fear and uncertainty. Therefore, the VIX Index can be considered an inverse leading indicator of the S&P 500 Index

Gold Prices - GOLD_US_Equity - Gold is often considered a safe haven asset, and its price tends to rise during times of economic uncertainty. Therefore, changes in the gold price may be able to predict changes in the S&P 500 Index, which is sensitive to economic conditions.

BTC – BTC_Index - Bitcoin is a relatively new asset class, but it has already shown a high degree of correlation with the S&P 500 Index. This suggests that Bitcoin may be able to provide some insight into the future performance of the U.S. stock market.

NIFTY 50 - NIFTY_Index - The NIFTY 50 is a benchmark Indian stock market index that represents the weighted average of 50 of the largest Indian companies listed on the National Stock Exchange. The NIFTY 50 Index is a broad representation of the Indian stock market, and as such, it can reflect global economic trends that may also impact the U.S. stock market.

RSI (Relative Strength Index) - The RSI is a momentum oscillator that measures the speed and magnitude of price movements in a stock or index. An RSI reading above 70 indicates that a stock or index is overbought, while a reading below 30 indicates that it is oversold. Overbought and oversold conditions often precede price reversals, so the RSI can be used as a potential indicator of future S&P 500 Index movements.

Fear and Greed Indicator- The Fear and Greed Indicator, developed by CNNMoney, combines seven different sentiment indicators to measure investor fear and greed. A higher Fear and Greed Index reading indicates that investors are more fearful, while a lower reading indicates that they are more greedy. Fearful investors tend to sell stocks, which can lead to lower prices, while greedy investors tend to buy stocks, which can lead to higher prices. Therefore, the Fear and Greed Indicator can be considered a potential indicator of future S&P 500 Index movements.

USD/JPY Currency Pair - USDJPY_ Currency - The USD/JPY currency pair is a measure of the relative strength of the U.S. dollar against the Japanese yen. A stronger dollar can make U.S. exports more expensive for foreign buyers, which can hurt corporate profits and weigh on stock

prices. Therefore, changes in the USD/JPY currency pair may be able to predict changes in the S&P 500 Index.

USD/INR Pair (U.S. Dollar/Indian Rupee Exchange Rate): The USD/INR currency pair measures the relative strength of the U.S. dollar against the Indian rupee. A stronger dollar can make U.S. exports more expensive for Indian buyers, which can hurt corporate profits and weigh on stock prices in India. Since the Indian economy is closely linked to the global economy, changes in the USD/INR currency pair may also have an indirect impact on the S&P 500 Index.

OIL Prices (Brent Crude Oil Futures): Oil is a major input for many industries, and its price can have a significant impact on corporate profits. When oil prices are high, it can lead to higher costs for businesses, which can eat into their profits and potentially lead to lower stock prices. Therefore, changes in oil prices can be considered a potential leading indicator of the S&P 500 Index.

Dow Jones U.S. Financial Services Index - DJUSFV_Index - The Dow Jones U.S. Financial Services Index tracks the performance of 20 of the largest and most influential financial services companies in the United States. The financial sector is a major driver of the U.S. economy, and the performance of the Dow Jones U.S. Financial Services Index can be considered a barometer of the overall health of the financial sector. Therefore, changes in the Dow Jones U.S. Financial Services Index can be considered a potential leading indicator of the S&P 500 Index.

Dow Jones U.S. Real Estate Index- DJUSRE_Index- The Dow Jones U.S. Real Estate Index tracks the performance of 30 of the largest and most influential real estate companies in the United States. The real estate sector is a significant part of the U.S. economy, and the performance of the Dow Jones U.S. Real Estate Index can be considered a barometer of the overall health of the real estate sector. Therefore, changes in the Dow Jones U.S. Real Estate Index can be considered a potential leading indicator of the S&P 500 Index.

Hang Seng Index - HSI Index-: The Hang Seng Index is a stock market index that measures the performance of the 50 largest companies listed on the Hong Kong Stock Exchange. It is a broad representation of the Hong Kong stock market, and as such, it can reflect global economic trends that may also impact the U.S. stock market. Therefore, the Hang Seng Index can be considered a leading indicator of the S&P 500 Index

Federal funds effective rate - FEDL01 Index- The federal funds effective rate serves as a crucial gauge for the overall health of the economy and acts as a key indicator for the future performance of the S&P 500 Index. Set by the Federal Open Market Committee (FOMC), the rate influences borrowing costs for businesses and consumers, directly impacting economic activity. When the FOMC hikes the federal funds rate, borrowing becomes more expensive, potentially dampening economic growth. Conversely, lowering the rate makes borrowing more affordable, stimulating economic expansion. Consequently, monitoring changes in the federal

funds rate provides valuable insights into future economic trends and their potential impact on corporate profits and stock prices.

Russell 2000 Index - RTY_Index - The Russell 2000 Index is a stock market index that measures the performance of the 2,000 smallest companies listed on the U.S. stock exchanges. It is considered to be a more volatile index than the S&P 500 Index, as smaller companies are generally more sensitive to changes in economic conditions. However, the Russell 2000 Index has also historically outperformed the S&P 500 Index over the long term. Therefore, the Russell 2000 Index can be considered a leading indicator of the S&P 500 Index, as its performance may signal future trends in the broader U.S. stock market.

United States Commodity Index Fund - USCI_US_Equity - The United States Commodity Index Fund (USCI) is a commodity pool that tracks the performance of a basket of commodities, including gold, silver, oil, and natural gas. Commodities are often considered to be a hedge against inflation, as their prices tend to rise when the general level of prices is also rising. Therefore, changes in the price of commodities may be able to predict changes in the S&P 500 Index, which is sensitive to inflation.

U.S. 10 Year Treasury- USGG10YR_Inde - The U.S. 10 Year Treasury is a bond issued by the U.S. Treasury Department with a maturity of 10 years. The yield on the U.S. 10 Year Treasury is considered to be a benchmark interest rate for the long-term cost of borrowing in the U.S. economy. When the yield on the U.S. 10 Year Treasury rises, it can make borrowing more expensive for businesses and consumers, which can slow down economic growth. Conversely, when the yield on the U.S. 10 Year Treasury falls, it makes borrowing more affordable, which can stimulate economic growth. Therefore, changes in the yield on the U.S. 10 Year Treasury can be considered a leading indicator of the S&P 500 Index, as they may signal changes in economic conditions that may affect corporate profits and stock prices.

APPLE - AAPL_US_Equity - Apple is one of the largest and most influential companies in the world, and its stock price is closely followed by investors around the globe. Therefore, changes in Apple's stock price may be able to predict changes in the broader U.S. stock market.

MICROSOFT- MSFT_US_Equity - Microsoft is another large and influential company with a global reach, and its stock price is also closely followed by investors. Therefore, changes in Microsoft's stock price may also be able to predict changes in the S&P 500 Index.

MOMENTUM - The momentum indicator is a financial metric used to measure the speed or velocity of price changes in a security. It captures the rate at which the price is moving up or down, serving as a tool to gauge the strength of a price trend. Typically, positive momentum indicates an upward trend in prices, suggesting increasing investor interest, while negative momentum signals a downward price trend, reflecting waning investor confidence.

MOM_MA - Represents the Moving Average of Momentum, a technical indicator that averages the momentum over a specified period, providing insights into the sustained strength of a stock or market index's movement, such as the S&P 500. A rising MOM_MA suggests increasing bullish momentum, while a falling MOM_MA indicates growing bearish momentum, reflecting shifts in market sentiment and trends.

10yr - 2yr -The yield spread between the 10-year and 2-year Treasury notes. The yield spread is a widely monitored indicator of the yield curve's steepness and is considered a barometer of economic sentiment. A positive spread suggests normal, healthy economic conditions, while a negative spread, or inversion, is often seen as a predictor of an economic downturn.

Gauss Markov assumption of the Model with significant factors:

Assumption 1:Linearity - The assumption of linearity appears to not hold in our model. The residuals do not scatter randomly around the horizontal line at zero but instead display a systematic pattern. This pattern suggests that the relationship between some of the predictors and the predicted variable may not be properly captured by the model.

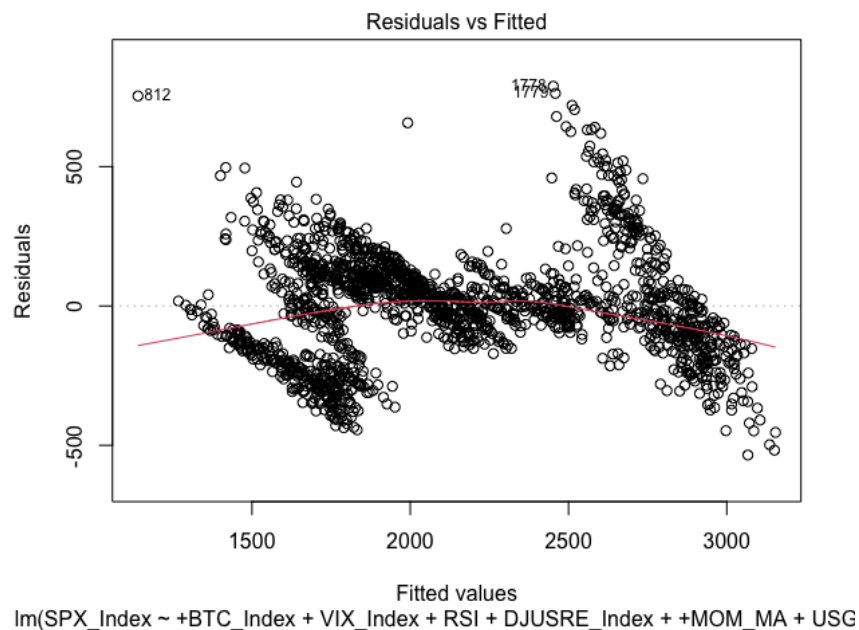


Fig 1. illustrates non-random residual patterns indicating potential non-linearity in the regression model

Assumption 2: Independence and Randomness - We run the Durbin-Watson Test to check for Independence and Randomness of error terms. It checks for autocorrelation with Lag 1. D-W statistics is 0.1339252 with a p-value of 0. This test result indicates strong evidence against the null hypothesis, suggesting the presence of positive autocorrelation. failing the assumption.

Null Hypothesis: $H_0: \rho = 0$

Alternative Hypothesis: $H_a: \rho > 0$

```
> durbinWatsonTest(spx_model_2, alternative = "positive")
lag Autocorrelation D-W Statistic p-value
1      0.9289984      0.1339252      0
Alternative hypothesis: rho > 0
```

Fig 2. illustrates a snapshot of the code of the Durbin-Watson Test

Assumption 3: Homoscedasticity - We are using the Breusch Pagan test to check if there is homoscedasticity. Consider, H_0 : There is no c.h. in the model, and H_1 : there is a c.h. in the model. With a Breuch Pagan test value of 307.22 and a p-value below $2.2e-16$ and $\alpha = 0.01$. We reject the null hypothesis suggesting that there is evidence of heteroskedasticity in the model based on the Breusch-Pagan test.

```
> bptest(spx_model_2)

studentized Breusch-Pagan test

data:  spx_model_2
BP = 307.22, df = 9, p-value < 2.2e-16
```

Fig . illustrates snapshot of the code of the Breusch Pagan Test

Assumption 4: Multicollinearity - We are using the Variance Inflation Factor (VIF) to see if there is significant multicollinearity in the model. All computed VIF values are below the critical value of 5, suggesting multicollinearity might be present in the model. But does not reach a level that would compromise the model's coefficient estimates. The highest VIF value, observed for ATR at 4.416480, approaches but does not go over the threshold of concern, implying that while correlations exist, they are unlikely to significantly affect the reliability of the regression results.

If $VIF = 1$, then it indicates the absence of multicollinearity.

If $VIF > 5$, multicollinearity might be present.

If $VIF > 10$, there is multicollinearity among the variables.

```
> #check for multicollinearity
> vif(spx_model_2)
```

BTC_Index	VIX_Index	RSI	DJUSRE_Index	MOM_MA	USGG10YR_Index
1.006228	3.457617	3.390583	1.076644	3.014514	1.331514
FEAR_GREED	ATR	`10yr-2yr`			
3.316582	4.416480	1.790842			

```
> |
```

Fig 3. illustrates snapshot of the code for VIF

Assumption 5: Zero Conditional Mean of Error Terms - The zero conditional mean of error terms is a crucial assumption in regression models because it indicates that the model should have an expected value of zero when explanatory variables are used. If the mean of the error terms is not zero, this could indicate bias or a failure to capture certain systematic information. The mean of the error terms in this case is $-4.54747e-14$, as seen in the Fig 4.below, which is close to zero, indicating that there is no systematic pattern that would violate the zero conditional mean assumption. This implies that the model has no apparent bias in its predictions based on the variables included, ensuring its predictive accuracy and the validity of statistical inferences drawn from it.

```
> mean(spx_model_2$residuals) #zero conditional mean
[1] -4.547474e-14
```

Fig 4. illustrates a snapshot of the code illustrating the zero-conditional mean obtained from the model

Assumption 6: Normality - When checking for linearity, the majority of the points appear to be normally distributed. Indicating the adherence to the normality assumption. But there are deviations at the tails, particularly in the upper tail, where the plot indicates that the residuals have heavier tails than expected under a normal distribution. While some deviation from the reference line is common, the extent of the deviation observed in our analysis suggests that the normality assumption may not hold, particularly impacting the tails of the residual distribution.

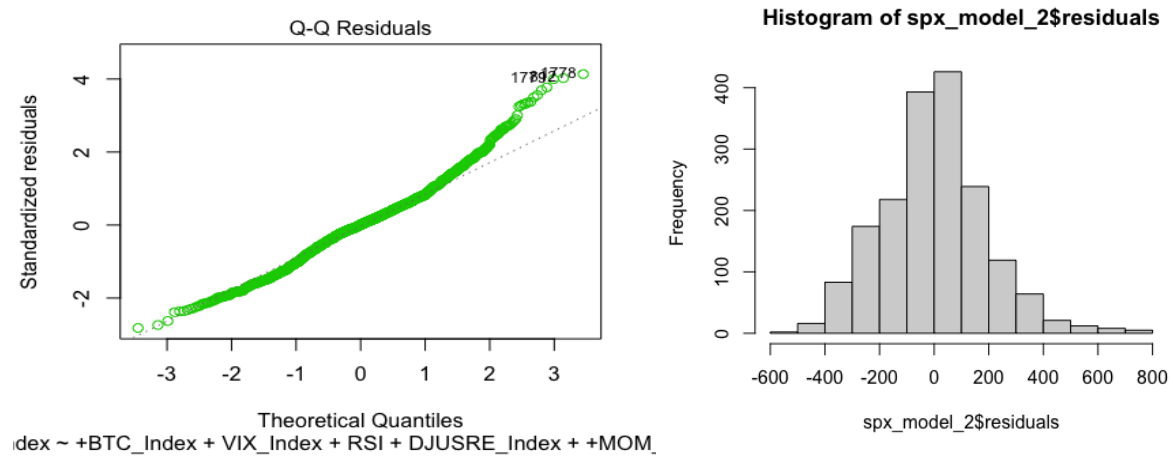


Fig 5. (left) illustrates the Quantile-Quantile (Q-Q) plot of Model residuals showing deviation from normality in the tail regions. Fig 6. (right) illustrates the histogram of Model with distribution of the residuals with a long tail

STATISTICAL ANALYSIS (HYPOTHESIS TESTING)

(I) Original model

```

Residuals:
    Min       1Q   Median       3Q      Max
-490.83 -109.15    7.69  106.42  758.34

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2798.4306   66.6244  42.003  < 2e-16 ***
NIFTY_Index    -1.0519    5.7732  -0.182  0.855441
GOLD_US_Equity  1.8854    1.7712   1.064  0.287276
BTC_Index     -1.1917    0.7238  -1.646  0.099864 .
USDJPY_Curncy -9.1429    8.7003  -1.051  0.293460
AAPL_US_Equity  3.1938    2.9898   1.068  0.285570
MSFT_US_Equity  0.8187    3.6442   0.225  0.822278
HIS_Index     -6.2476    4.7084  -1.327  0.184710
FEDL01_Index  117.4848   17.2508   6.810  1.33e-11 ***
USDINR_Curncy -4.1799   10.3167  -0.405  0.685409
RTY_Index      0.4145    7.9774   0.052  0.958569
DJUSFV_Index  -10.1413    7.0237  -1.444  0.148956
VIX_Index     -60.1629    2.1866 -27.514  < 2e-16 ***
RSI            -3.0113    0.7853  -3.835  0.000130 ***
MOMENTUM      -0.7178    0.2117  -3.390  0.000714 ***
MOM_MA         0.9335    0.2214   4.217  2.60e-05 ***
USCI_US_Equity -9.2380    8.5537  -1.080  0.280293
DJUSRE_Index  -14.8840    6.1354  -2.426  0.015370 *
USGG10YR_Index 224.3273   16.3889  13.688  < 2e-16 ***
FEAR_GREED     0.7017    0.1767   3.971  7.45e-05 ***
ATR            14.1636    0.8052  17.589  < 2e-16 ***
`10yr-2yr`    -378.1597   19.3196 -19.574  < 2e-16 ***
OIL_Price      -0.2809    2.4788  -0.113  0.909795
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 187.3 on 1757 degrees of freedom
Multiple R-squared:  0.8624,    Adjusted R-squared:  0.8607
F-statistic: 500.8 on 22 and 1757 DF,  p-value: < 2.2e-16

```

Fig 7. illustrates a snapshot of the code representing the original model

The Fig 7. shows the results of a regression analysis of the original model with the 22 factors to estimate the S&P 500 Index. The model has a highly significant F-statistic and an R^2 value of 0.8624, suggesting a 86.24% of variance of the S&P 500 Index. The coefficients of this model highlights the estimates, standard errors, significance levels for each predictor, where some variables show strong statistical significance with the movement of the index. When all the explanatory variables are set to zero the S&P 500 Index will be at 2798.4306. Fig7. displays all the parameter values. Five of the parameters have a negative effect on the S&P 500 Index.

We are going to check if the model's explanatory variables are jointly significant. Consider $H_0: B_i = 0, i = 1, 2, \dots, 22$ and $H_1: B_i \neq 0, i = 1, 2, \dots, 22$ at $\alpha = 0.1$. The model has an F-statistics of 500.8

with a p-value of less than $2.2e-16$, which is well below the significance level of 0.01. Hence we reject H_0 . There is enough evidence the explanatory variables are jointly significant in the model. When we check if the explanatory variables are individually significant. Consider, $H_0: B_i = 0$, $i = 1, 2, \dots, 22$ and $H_1: B_i \neq 0$, $i = 1, 2, \dots, 22$ at $\alpha = 0.1$. We identify that the explanatory variables NIFTY Index, Gold, AAPL, MSFT, HIS Index, USDNIR, RTY Index, DJUSFV Index, USCI, and Oil are insignificant.

```

Residuals:
    Min       1Q   Median       3Q      Max
-490.83 -109.15    7.69  106.42  758.34

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2798.4306   66.6244  42.003  < 2e-16 ***
NIFTY_Index   -1.0519    5.7732  -0.182  0.855441
GOLD_US_Equity  1.8854    1.7712   1.064  0.287276
BTC_Index     -1.1917    0.7238  -1.646  0.099864 .
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RSI            -3.0113    0.7853  -3.835  0.000130 ***
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DJUSRE_Index  -14.8840    6.1354  -2.426  0.015370 *
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FEAR_GREED     0.7017    0.1767   3.971  7.45e-05 ***
ATR            14.1636    0.8052  17.589  < 2e-16 ***
`10yr-2yr`    -378.1597   19.3196 -19.574  < 2e-16 ***
OIL_Price      -0.2809    2.4788  -0.113  0.909795
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 187.3 on 1757 degrees of freedom
Multiple R-squared:  0.8624,    Adjusted R-squared:  0.8607
F-statistic: 500.8 on 22 and 1757 DF,  p-value: < 2.2e-16

```

Fig 7. illustrates a snapshot of the code representing the original model

(II) Model with only significant factors

The Fig 8. above shows the results of the regression analysis of a model, where all the insignificant factors are deleted. We also exclude Momentum and FEDL01 Index due to high correlation to other factors. The model has a multiple R^2 of 0.8548, suggesting a 85.48% of

variance of the S&P 500 Index. When all the explanatory variables are set to zero the S&P 500 Index will be at 2878.2010. And displays all the parameter values. Five of the parameters have a negative effect on the S&P 500 Index.

We are going to check if the model's explanatory variables are jointly significant. Consider $H_0: B_i = 0, i = 1, 2, \dots, 9$ and $H_1: B_i \neq 0, i = 1, 2, \dots, 9$ at $\alpha = 0.01$. The model has an F-statistics of 1165 with a p-value of less than $2.2e-16$, which is well below the significance level of 0.01. Hence we reject H_0 . There is enough evidence the explanatory variables are jointly significant in the model. When we check if the explanatory variables are individually significant. Consider $H_0: B_i = 0, i = 1, 2, \dots, 9$ and $H_1: B_i \neq 0, i = 1, 2, \dots, 9$ at $\alpha = 0.01$. We identify that the explanatory variable BTC Index is insignificant.

```

Residuals:
    Min       1Q   Median       3Q      Max
-534.68 -116.84    1.67  106.61  788.31

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2878.2010    66.7651  43.109  < 2e-16 ***
BTC_Index     -0.9998     0.7355  -1.359  0.174201
VIX_Index    -58.9858     2.2078 -26.717  < 2e-16 ***
RSI           -3.9145     0.7606  -5.147  2.95e-07 ***
DJUSRE_Index -23.0396     5.1549  -4.469  8.34e-06 ***
MOM_MA         0.6164     0.1704   3.618  0.000305 ***
USGG10YR_Index 307.5052    12.0875  25.440  < 2e-16 ***
FEAR_GREED     0.4726     0.1654   2.858  0.004319 **
ATR           13.9057     0.8027  17.323  < 2e-16 ***
`10yr-2yr`    -500.4222    8.7668 -57.082  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 191.2 on 1770 degrees of freedom
Multiple R-squared:  0.8556,    Adjusted R-squared:  0.8548
F-statistic: 1165 on 9 and 1770 DF,  p-value: < 2.2e-16

```

Fig 8. illustrates a snapshot of the code representing the model with only significant factors

VIOLATIONS AND RECOMMENDATIONS

Several identified violations of the assumptions of linearity, independence and randomness, homoskedasticity, and normality emerged during the regression analysis. Subsequently, specific recommendations are provided to tackle these concerns in order to improve the model's resilience and accuracy in estimation which is illustrated in Table 1.

Violations of Assumptions	Recommendations
Linearity	Transform independent variables (logarithmic, percent change, etc), adding/removing independent variables through trial and error
Independence and Randomness	Investigate omitted variables or use generalized difference equations (Cochrane-Orcutt or AR(1) methods)
Homoskedasticity	Weighted or generalized least squares should be used due to autocorrelation. If autocorrelation is eliminated, weighted least squares would be more suitable.
Normality	Normality may be achieved by transforming the dependent variable with a logarithmic or square root operation. Additionally, outliers can be eliminated from regression analysis.

Table 1. illustrating the suggested recommendations respective to the assumption violations.

CONCLUSION

By employing multivariable linear regression, we have identified several key economic and financial indicators that have significant influence over stock market performance. These factors are Bitcoin Index, VIX Index, RSI, DJUSRE Index, MOM MA, USGG10YR Index, Fear Greed, ATR, and 10yr - 2yr. The quality of our model was ensured through adhering to Gauss Markov assumptions.

The Regression equation is:

$$\begin{aligned} \text{S\&P 500 Index} = & 0.9998 * \text{BTC_Index} - 58.9588 * \text{VIX_Index} - 3.9145 * \text{RSI} - \\ & 23.0396 * \text{DJUSRE_Index} + 0.6164 * \text{MOM_MA} + 307.5052 * \text{USGG10YR_Index} + \\ & 0.4726 * \text{FEAR_GREED} + 13.9057 * \text{ATR} - 500.4222 * \text{10yr-2yr} \end{aligned}$$

CITATION AND REFERENCES

VIX Index

- Bloomberg Terminal. Accessed 2023. Chicago Mercantile Exchange. "CBOE Volatility Index (VIX)." CBOE, 2023, www.cboe.com/tradable_products/vix/.

GOLD_US_Equity

- Bloomberg Terminal Data: Access the XAUUSD (Gold/US Dollar) currency pair on the Bloomberg Terminal.
- Reference: World Gold Council. (2023). Gold Prices. <https://www.gold.org/goldhub/data/gold-prices>

BTC_Index

- Bloomberg Terminal Data: Access the BTCUSD (Bitcoin/US Dollar) currency pair on the Bloomberg Terminal.
- Reference: CoinMarketCap. (2023). Bitcoin (BTC) Price Charts & Market Cap. <https://coinmarketcap.com/currencies/bitcoin/>

NIFTY_Index

- Bloomberg Terminal Data: Access the NIFTY50 (NIFTY 50 Index) on the Bloomberg Terminal.
- Reference: National Stock Exchange of India. (2023). NIFTY 50 Index. <https://www.nseindia.com/market-data/live-market-indices>

RSI

- Bloomberg Terminal Data: Access the RSI technical indicator for the desired market instrument.
- Reference: Wilder, J. Welles. (1978). New Concepts in Technical Trading Systems. Trend Research.

USD/JPY Currency Pair

- Bloomberg Terminal Data: Access the USDJPY (USD/JPY) currency pair on the Bloomberg Terminal.

- External Reference: Federal Reserve Bank of St. Louis. (2023). Euro Currency Exchange Rate. <https://fred.stlouisfed.org/categories/15>

OIL Prices

- Bloomberg Terminal Data: Access various crude oil futures contracts like WTI (West Texas Intermediate) or Brent on the Bloomberg Terminal.
- External Reference: U.S. Energy Information Administration. (2023). Petroleum & Other Liquids. <https://www.eia.gov/petroleum/>

DJUSFV_Index

- Bloomberg Terminal Data: Access the DJUSFV (Dow Jones U.S. Financials Index) on the Bloomberg Terminal.
- External Reference: S&P Global Dow Jones Indices. (2023). Dow Jones U.S. Financials Index. <https://www.spglobal.com/spdji/en/indices/equity/sp-500-financials-sector/>

DJUSRE_Index

- Bloomberg Terminal Data: Access the DJUSRE (Dow Jones U.S. Real Estate Index) on the Bloomberg Terminal.
- External Reference: S&P Global Dow Jones Indices. (2023). Dow Jones U.S. Real Estate Index. <https://www.spglobal.com/spdji/en/index-family/equity/property-reit/sp-property/>

HSI Index

- Bloomberg Terminal Data: Access the HSI (Hang Seng Index) on the Bloomberg Terminal.
- External Reference: Hang Seng Indexes Company Limited. (2023). Hang Seng Index. <https://www.hsi.com.hk/>

10yr - 2yr

- Bloomberg Terminal Data: Access the spread between the US 10-Year Treasury Bond (UST10) and the US 2-Year Treasury Bond (UST2).

- Reference: U.S. Department of the Treasury. (2023). Treasury Yields.
https://home.treasury.gov/resource-center/data-chart-center/interest-rates/TextView?type=daily_treasury_yield_curve&field_tdr_date_value_month=202202

AAPL_US_Equity

- Bloomberg Terminal Data: Access the AAPL (Apple Inc.) equity price.
- Reference: Apple Inc. (2023). Investor Relations.
<https://investor.apple.com/investor-relations/default.aspx>

MSFT_US_Equity

- Bloomberg Terminal Data: Access the MSFT (Microsoft Corporation) equity price.
- Reference: Microsoft Corporation. (2023). Investor Relations.
<https://www.microsoft.com/en-us/investor>

USCI_US_Equity

- Bloomberg Terminal Data: Access the USCI (S&P US Consumer Staples Select Sector Index) index.
- Reference: S&P Global Dow Jones Indices. (2023). Dow Jones U.S. Consumer Staples Select Sector Index.
<https://www.spglobal.com/spdji/en/indices/equity/sp-global-1200-consumer-staples-selector-capped-index/>

RTY_Index

- Bloomberg Terminal Data: Access the RTY (Russell 2000 Index) index.
- Reference: FTSE Russell. (2023). Russell 2000 Index.
<https://www.lseg.com/en/ftse-russell/indices/russell-us>

FEDL01 Index

- Bloomberg Terminal Data: Access the FEDL01 (Federal Reserve Bank of New York Discount Rate) index.
- Reference: Federal Reserve Bank of New York. (2023). Discount Rate.
<https://www.statista.com/statistics/187628/federal-reserve-bank-of-new-york-discount-rate-since-1990/>

USD/INR Pair (U.S. Dollar/Indian Rupee Exchange Rate)

- Bloomberg Terminal Data: Access the USDINR (USD/INR) currency pair.
- External Reference: Federal Reserve Bank of St. Louis. (2023). Euro Currency Exchange Rate. <https://fred.stlouisfed.org/categories/15>

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- Fear & Greed Index accessed through Bloomberg Terminal data on 2023-12-02.
- External Reference: Alternative.me. (n.d.). Fear & Greed Index. Retrieved from <https://alternative.me/crypto/>; <https://alternative.me/crypto/>

MOMENTUM Indicator

- Reference: Technical Indicators. (2023). Momentum (MOM) Indicator. Retrieved from <https://www.tradingview.com/scripts/momentum/>
- Optional Reference: GoodCrypto. (2023). An In-Depth Guide to Momentum Indicator (MOM). Retrieved from <https://goodcrypto.app/>

MOM_MA Indicator

- Reference: Warrior Trading. (n.d.). Momentum Indicator (MOM) Explained For Beginners. Retrieved from <https://www.warriortrading.com/momentum-trading-strategies/>
- Optional Reference: TradingView. (2023). Momentum Indicator (MOM) — Technical Indicators. Retrieved from <https://www.tradingview.com/scripts/momentum/>

APPENDIX

```

3 library(dplyr)
4 library(ggplot2)
5 library(readxl)
6 library(lmtest)
7 library(car)
8 library(ggcorrplot)
9 library(memisc)
10
11 #read excel file and remove missing values
12 data_file <- read_excel("~/Desktop/MQF Fall Semester/Econometrics/Final Project /Final Project Data.xlsx",
13                       sheet = "Sheet1", na = " ") %>% na.omit(data_file)
14
15 #data from 01-02-2012 to 12-31-2019
16 data_file <- data_file[1:1781,]
17 head(data_file)
18
19 #convert index's to percent change
20 model_data <- data_file %>% mutate(across(.cols = c("NIFTY_Index", "GOLD_US_Equity", "BTC_Index", "USDJPY_Currency", "AAPL_US_Equity", "MSFT",
21 head(model_data)
22
23 #remove date column and first row with NA values
24 model_data <- model_data[-1,-1]
25 head(model_data)
26
27 #run the regression
28 spx_model <- lm(data = model_data, SPX_Index ~.)
29 summary(spx_model)
30
31 #check for multicollinearity
32 vif(spx_model)
33
34 #bp and dw tests
35 bptest(spx_model)
36 durbinWatsonTest(spx_model, alternative = "positive")
37
38 #residual analysis
39 plot(spx_model, 1) #residuals vs fitted
40 plot(spx_model, which = 2, col = c("blue")) #normal probability plot
41 plot(residuals(spx_model)) #residuals
42 hist(spx_model$residuals) #histogram
43
44 #-----
45 #run the regression with only significant factor
46 spx_model_2 <- lm(data = model_data, SPX_Index ~ + BTC_Index + VIX_Index + RSI + DJUSRE_Index +
47               + MOM_MA + USGG10YR_Index + FEAR_GREED + ATR + `10yr-2yr`)
48 summary(spx_model_2)
49
50 #check for multicollinearity
51 vif(spx_model_2)
52
53 #bp and dw tests
54 bptest(spx_model_2)
55 durbinWatsonTest(spx_model_2, alternative = "positive")
56
57 #residual analysis
58 plot(spx_model_2, 1) #residuals vs fitted
59 plot(spx_model_2, which = 2, col = c("green3")) #normal probability plot
60 plot(residuals(spx_model_2)) #residuals
61 hist(spx_model_2$residuals) #histogram
62 mean(spx_model_2$residuals) #zero conditional mean
63
64 #comparing models
65 model_comparison <- mtable(spx_model, spx_model_2)
66 model_comparison
67
68

```